An older input through the actual resultant

Lean proves that a surviving old variable's degree d becomes exactly $2d$ in the actual fixed-size Semaev resultant. The coefficient at $2d$ is the same resultant on the previous old leading coefficient. This supplies a canonical third-newest-input degree theorem at every arity at least five. [pcc:unverified]

Let $f \in S[\text{old}][E]$ be nonzero, with elimination degree at most the declared size D, and let the four curve coefficients belong to the domain S. The actual variable permutation reads $S[\text{old}][Z][T]$ as $S[Z][T][\text{old}]$. The Sylvester sizes remain $D, 2$ throughout. [pcc:unverified]

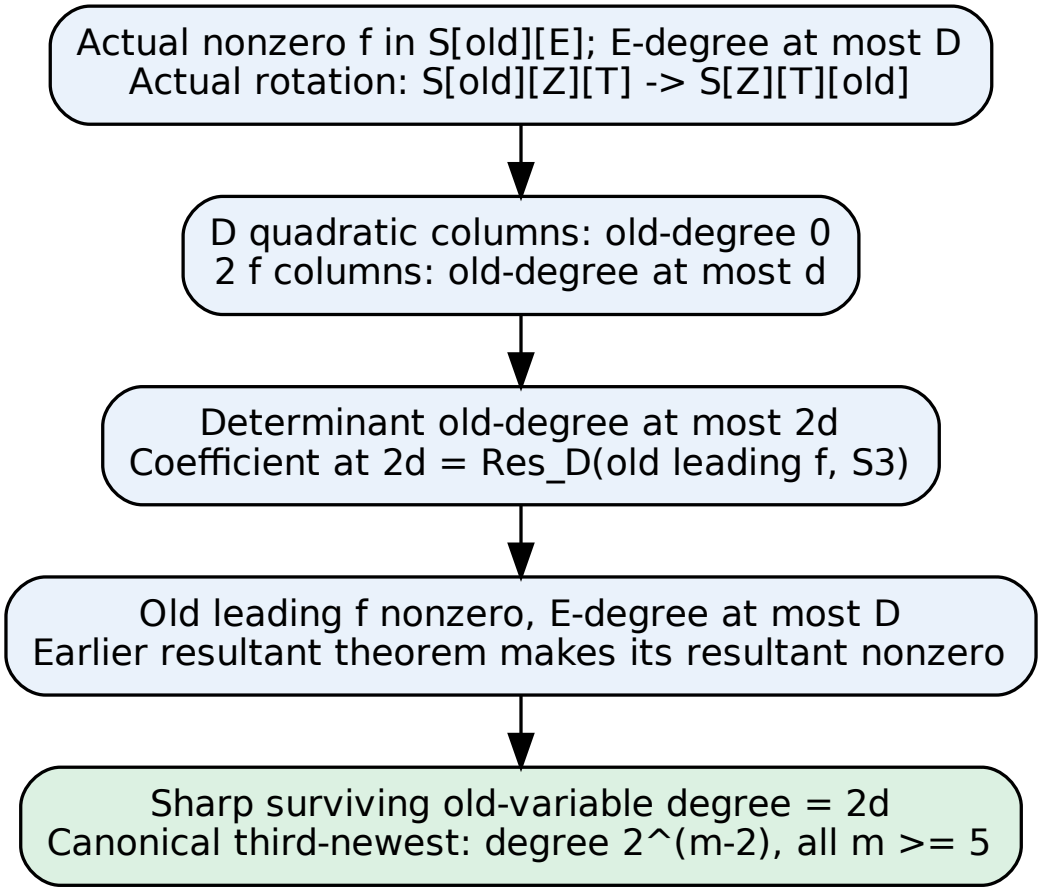


Figure 1: Actual determinant proof: the coefficient at the degree bound is another actual resultant, whose nonvanishing was proved in the earlier general step. [pcc:unverified]

The canonical consequence and its limits

At actual recursive stage $n+2$, arity $m = n+5$, the surviving input was the preceding polynomial's penultimate input. Its degree 2^{n+2} doubles to $2^{n+3} = 2^{m-2}$. The earlier newest-pair proof supplies the other two newest input degrees. [pcc:unverified]

Canonical arity	Checked individual input degrees
4	All four equal 4, from the earlier S4 proof
5	Three newest equal 8
Every $m \geq 5$	Three newest equal 2^{m-2}

The matrix has D quadratic columns of old degree zero and two f columns of old degree at most d. The determinant degree is at most $2d$. Its coefficient at $2d$ is the fixed-size resultant of the old leading polynomial with the lifted S3 quadratic. That old leading polynomial is nonzero and still has elimination degree at most D. The earlier exact resultant coefficient theorem proves this coefficient cannot vanish. No generic leading-term assumption is needed. [pcc:unverified]

Main binding: `focus_degree_audit_certificate` in RecursiveSemaevFocusAudit. Generic sharp degree and exact top coefficient are fully quantified alongside the actual canonical corollary. Proof map: `../../RECURSIVE_FOCUS.md`. Exact source/statement hashes and gate receipt: `../../recursive-focus-sources.json`. Whole census gate357passes, zero failures/zero trusted-native exceptions. Source commit `a55402b8` precedes packet `PCC-20261110-2c396d`. Actual local verdict and unsigned standing are in the separate verification record. [pcc:unverified]

Lean4.35.0 and Mathlib `56a00585` are pinned. Source15 compiles without warnings; only propext, Classical.choice and Quot.sound are used. Fourteen earlier attempts are preserved losslessly, including accepted intermediate components. Explicit polynomial hom domains/codomains resolve nested ring inference. No hole, custom axiom, native decision exception or linter waiver. [pcc:unverified]

Primary read scopes remain Kosters–Yeo2015 section2/Proposition2.1and related Semaev notes. Mathlib Bivariate.swap, Resultant.Basic and Matrix.Polynomial were reviewed. Original Semaev2004remains unread after 403; no novelty or source-status promotion. [pcc:unverified]

Every remaining older canonical input still needs transport through the successive coefficient-ring permutations. Full permutation symmetry, higher geometric zero-sum recursion and irreducibility remain open. No degree preservation after finite-field evaluation, new native experiment, timing or paid compute is claimed. [pcc:unverified]